

System Design

Master Problem List

32
problems

32 must-solve problems · 8 categories · interview-ready

INFRASTRUCTURE & CORE

- 01 Design a URL shortener
- 02 Design a web crawler
- 03 Design a rate limiter
- 04 Design a distributed cache (Redis)
- 05 Design a content delivery network (CDN)
- 06 Design a distributed job scheduler
- 07 Design a logging and metrics pipeline

COMMUNICATION & FEEDS

- 08 Design a notification system
- 09 Design a chat app (WhatsApp)
- 10 Design Slack
- 11 Design a news feed (Facebook / Twitter)

SOCIAL MEDIA

- 12 Design Twitter / X
- 13 Design Instagram
- 14 Design TikTok
- 15 Design Pinterest
- 16 Design Reddit

VIDEO & MEDIA STREAMING

- 17 Design YouTube
- 18 Design Netflix
- 19 Design Spotify

LOCATION & RIDE-SHARING

- 20 Design Uber
- 21 Design a food delivery app (Swiggy / Zomato)
- 22 Design Google Maps

BOOKING & INVENTORY

- 23 Design a hotel booking system (Booking.com)
- 24 Design a flight booking system
- 25 Design a movie ticket booking (BookMyShow)
- 26 Design a restaurant reservation system

STORAGE & SEARCH

- 27 Design Google Drive / Dropbox
- 28 Design Pastebin
- 29 Design a search autocomplete system
- 30 Design a trending topics system

E-COMMERCE & PAYMENTS

- 31 Design an e-commerce platform (Amazon)
- 32 Design a payment system / wallet (PhonePe)

ANCHOR PROBLEMS: Uber · Hotel Booking · Instagram · Twitter · YouTube · WhatsApp · Autocomplete

Master these 7 first — they unlock all 32 problems via shared patterns